SOLAR BATTERY CHARGER

Simulation demonstration and results

Open START_HERE.html in Microsoft Edge, Chrome, or Firefox for the interactive viewer.

Seven completed ngspice transient runs; 40 numerical checks passed; 0 failed.

Educational averaged behavioral equivalent, NOT the manufacturer LT3652 macromodel.

This demonstrates modeled behavior; it does not prove the physical PCB works.

Design targets: 4.20 V / 500 mA, 0.20 ohm sense resistor, nominal 12 V solar panel.

Demonstration assumptions: 2.5 Ah model cell; 0.15 ohm cell resistance; fixed 88% efficiency.

Time scaling: 1 SPICE second = 1 modeled charging hour. Real charging time is not validated.

Demonstrated conditions

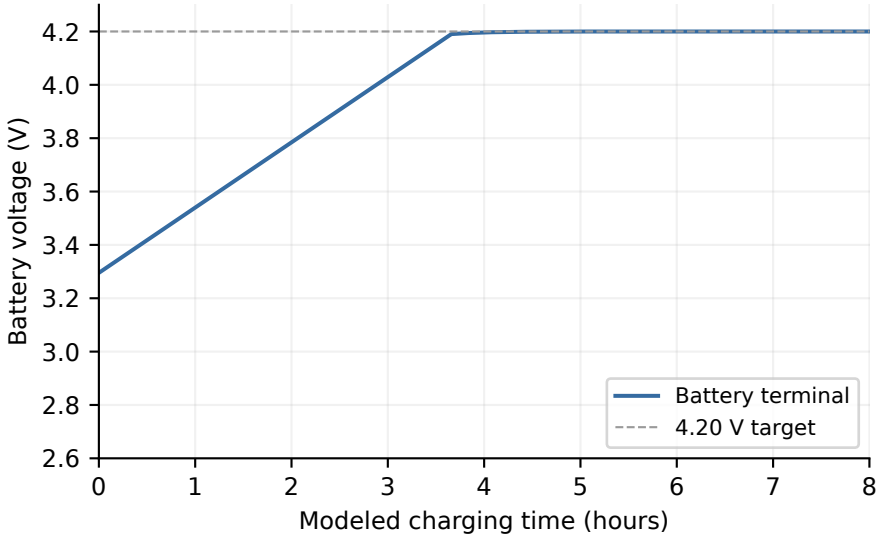
Normal charge / taper / finish · Hot and cold faults · Cloud / night / sunrise
Bad-battery timeout · Preconditioning · Load / recharge · Disconnected sensor

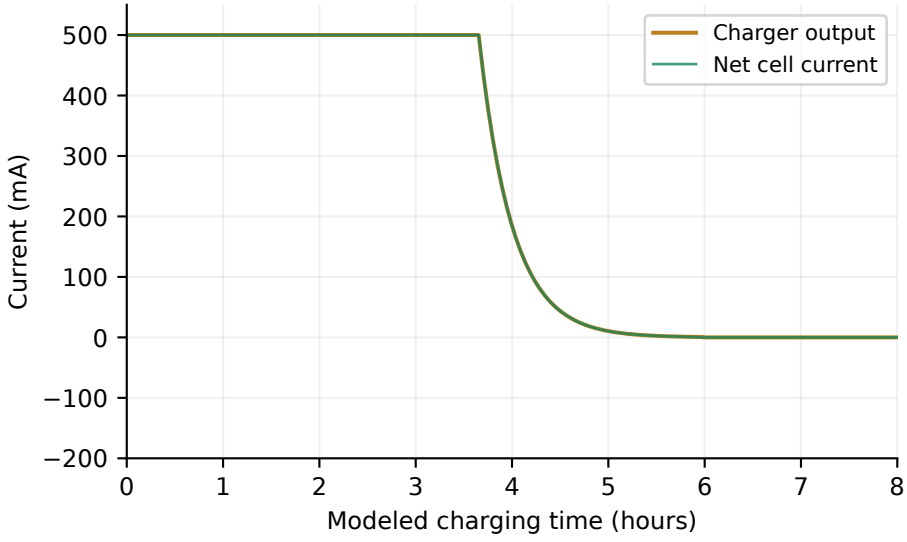
Read README.md for reproduction steps, modeling assumptions, and limitations.

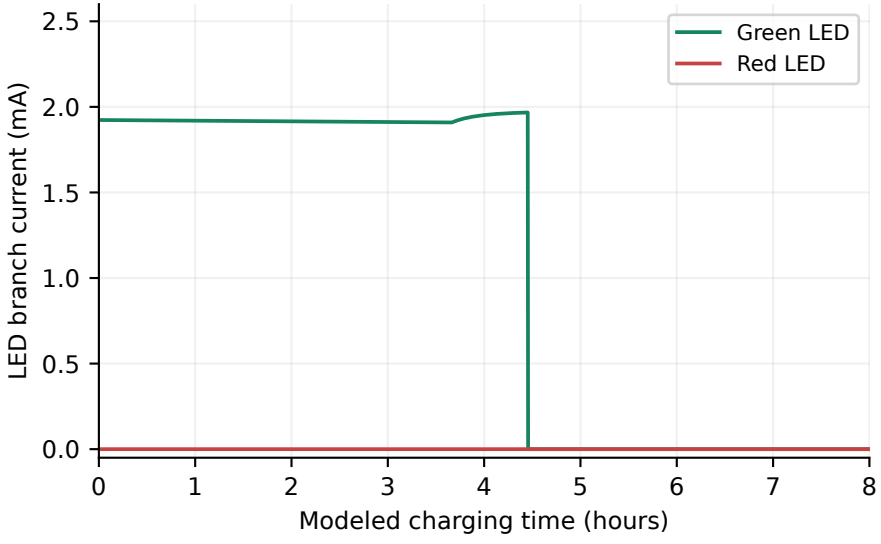
01 Charge, taper, and finish

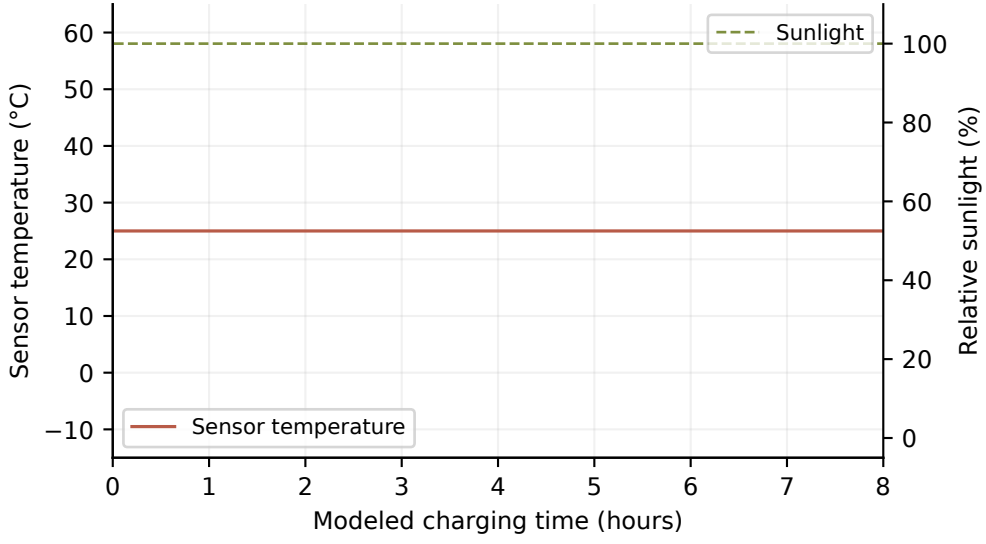
Bright sunlight: charge at 500 mA, approach 4.2 V, taper below the green-LED threshold, then reach timer termination.

EDUCATIONAL BEHAVIORAL MODEL — not an LT3652 manufacturer-model or PCB validation







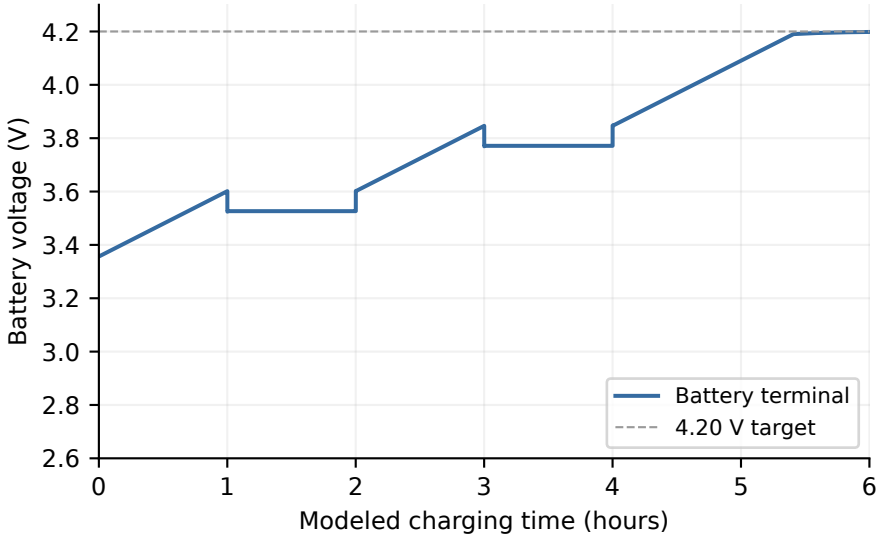


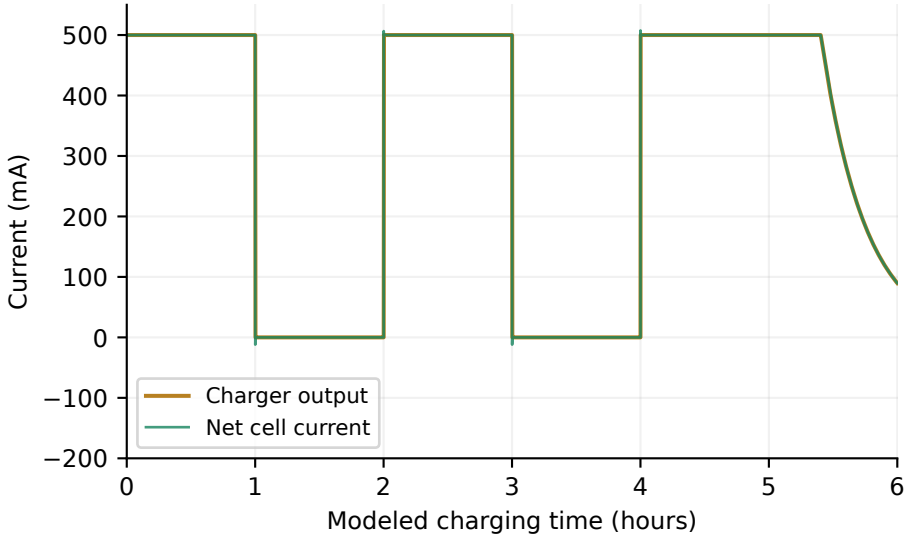
1 SPICE second = 1 modeled hour. Hour-scale plots; startup and fast transients are not validated. Full data: results/*.csv.

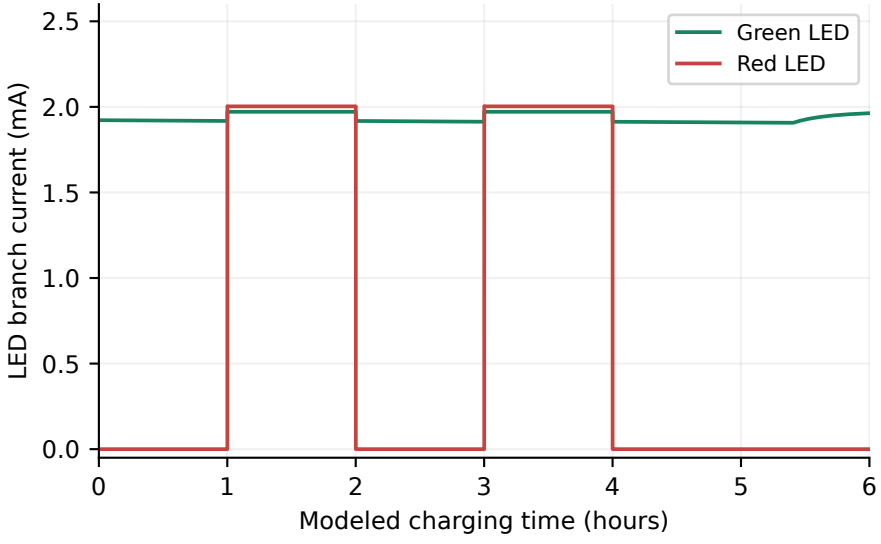
02 Hot / cold fault and recovery

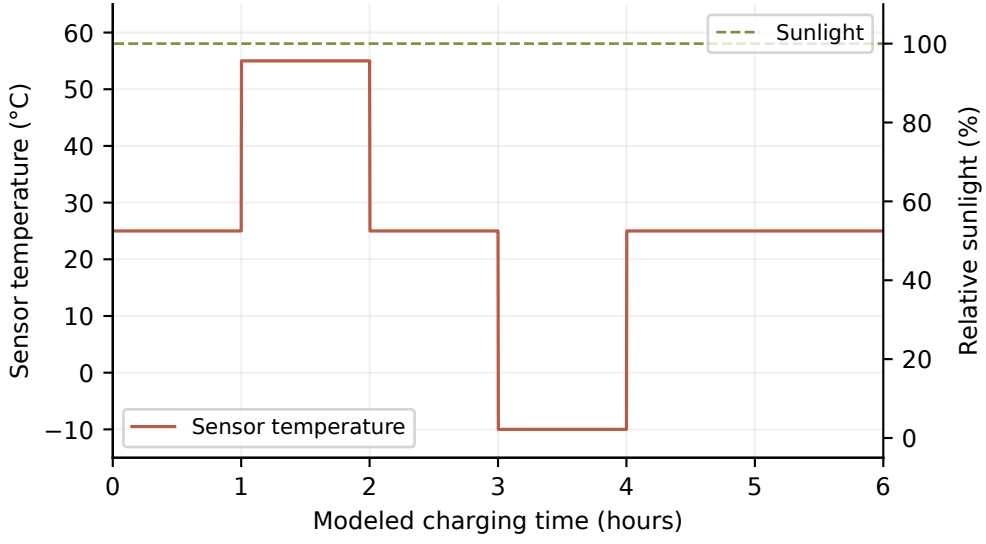
55 C and -10 C sensor conditions pause charging and light both LEDs. Returning to 25 C resumes charging.

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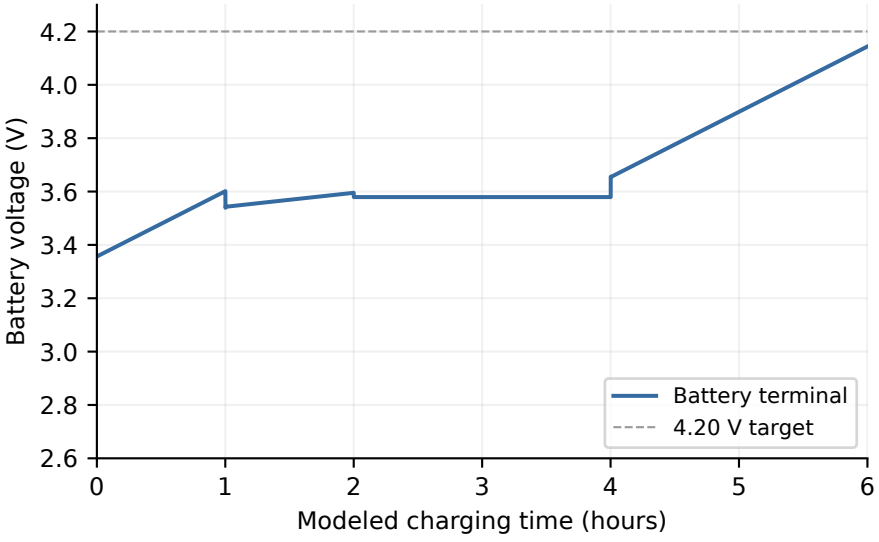


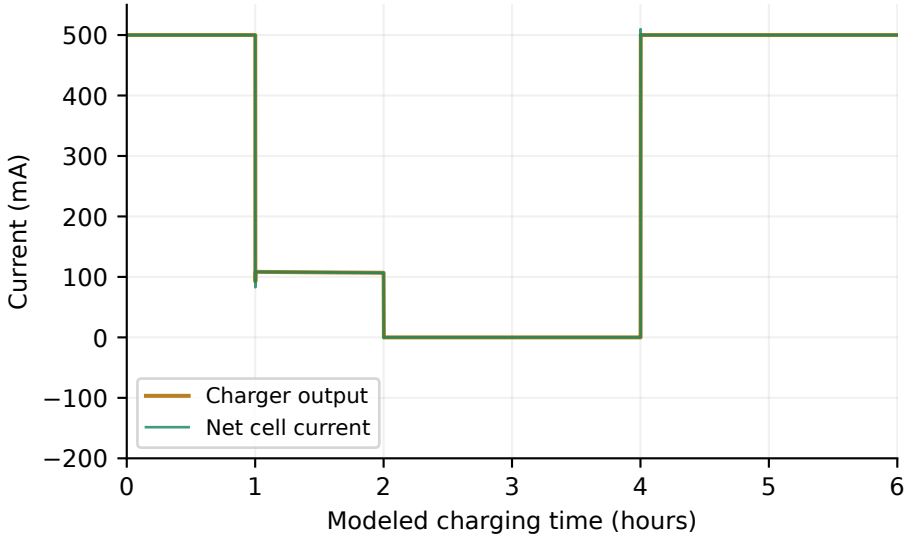
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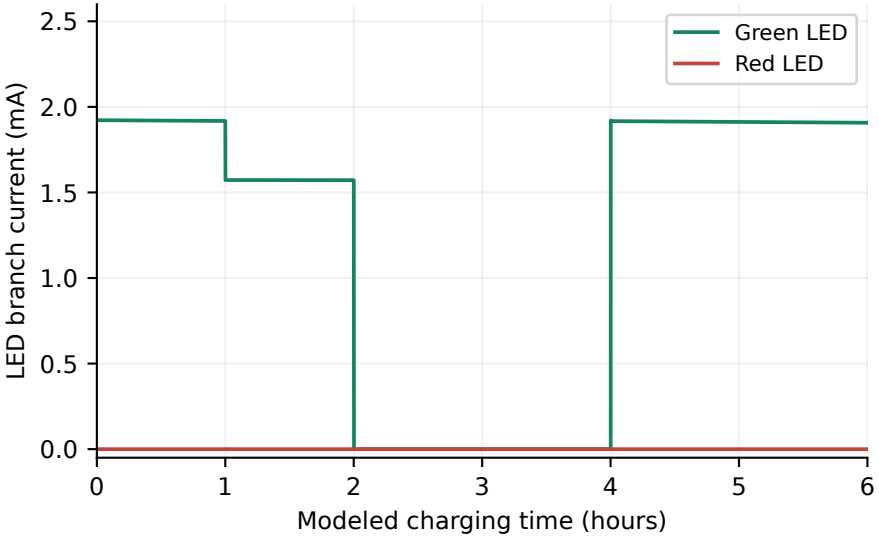
03 Clouds, night, and sunrise

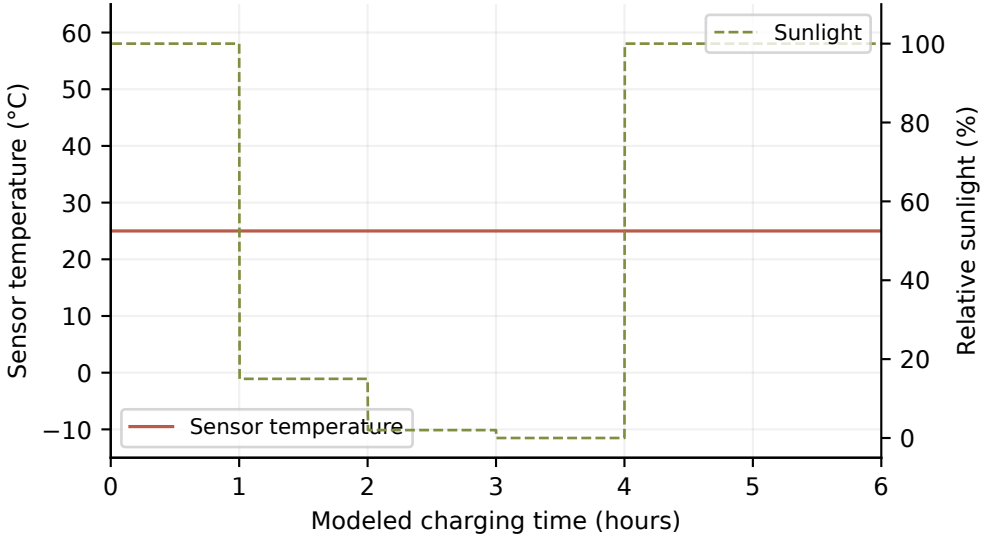
Partial cloud reduces available charging current. Very weak light can stop charging while the battery is not full. Night turns the solar-powered LEDs off.

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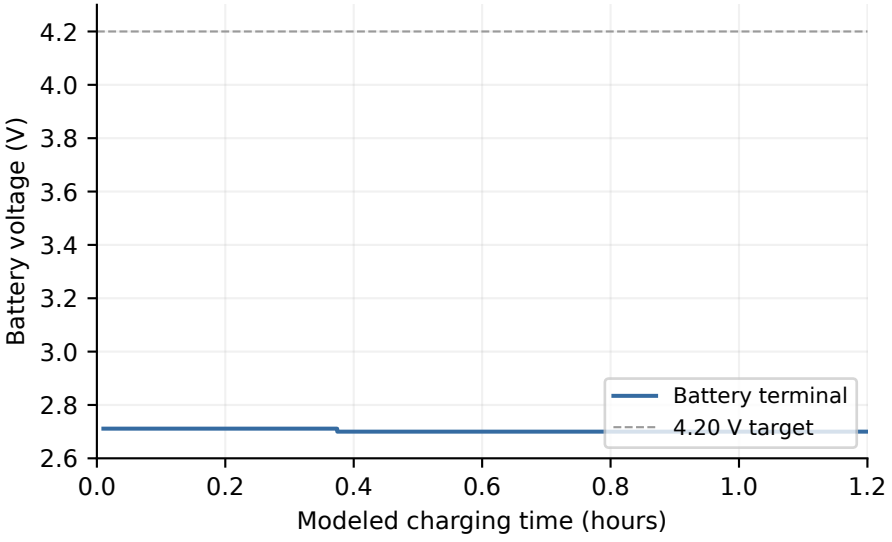


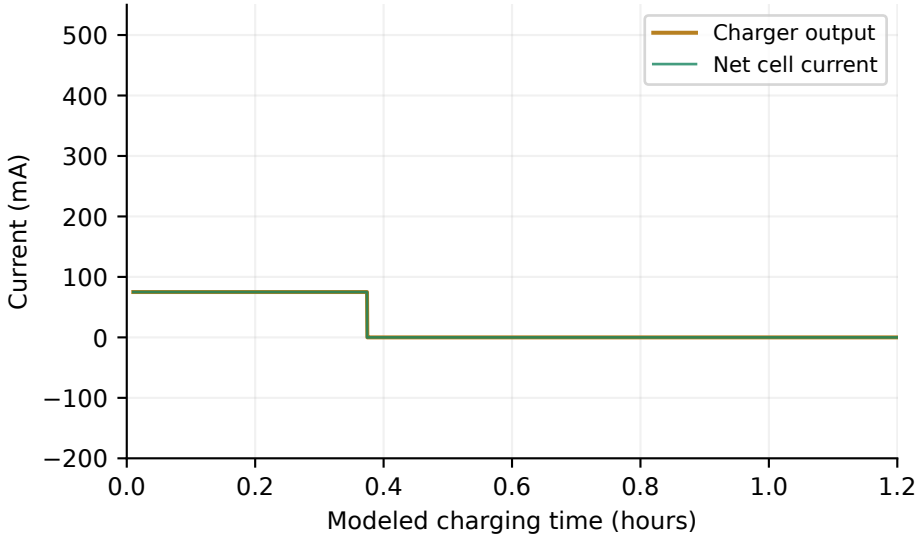
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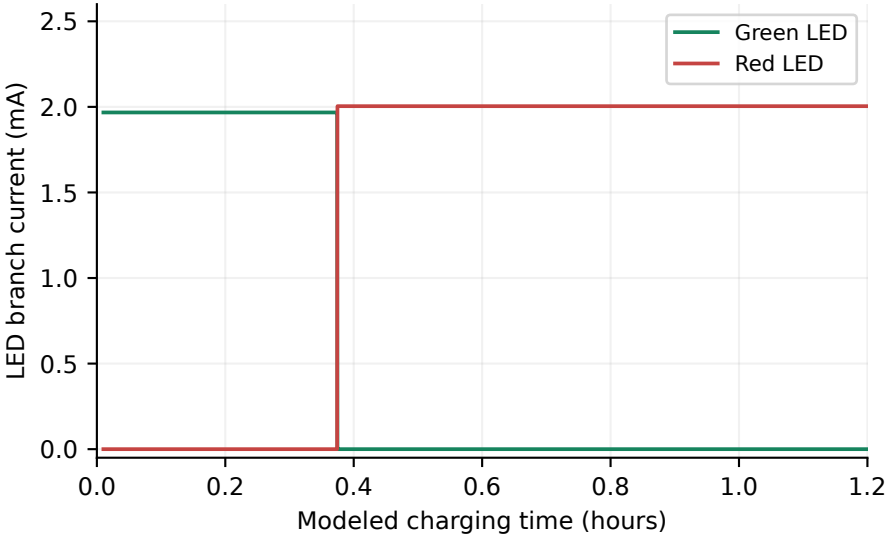
04 Bad-battery timeout

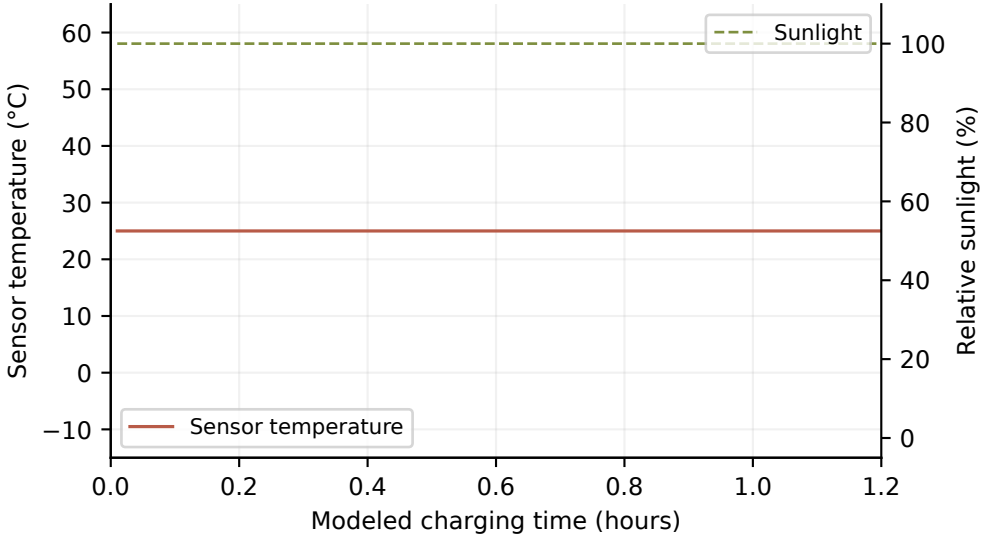
Fault-injection fixture: the cell voltage does not rise despite preconditioning current. After about 22.4 minutes the red LED turns on and charging stops.

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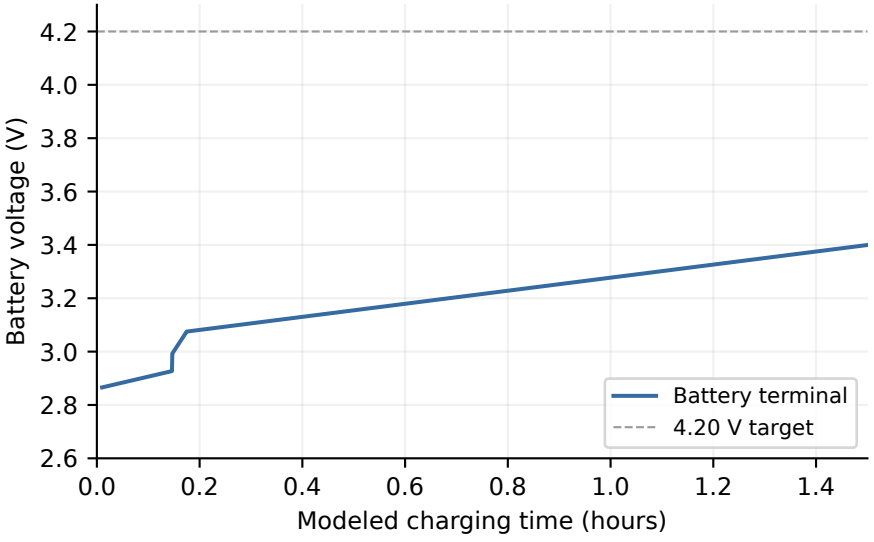


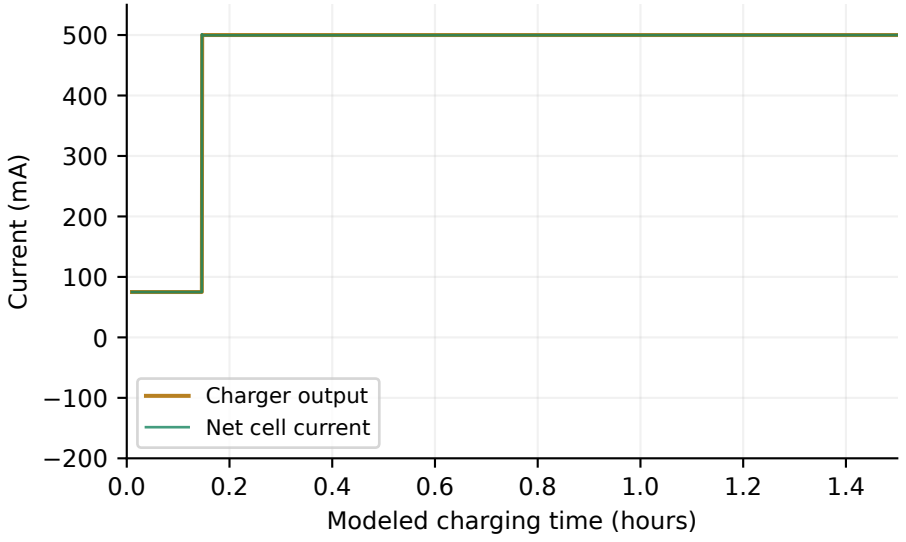
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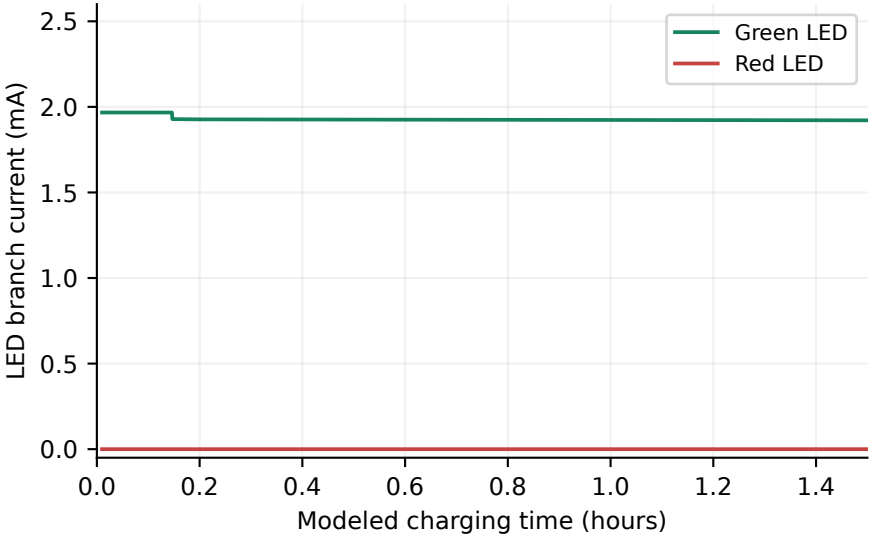
05 Recover a low-voltage model cell

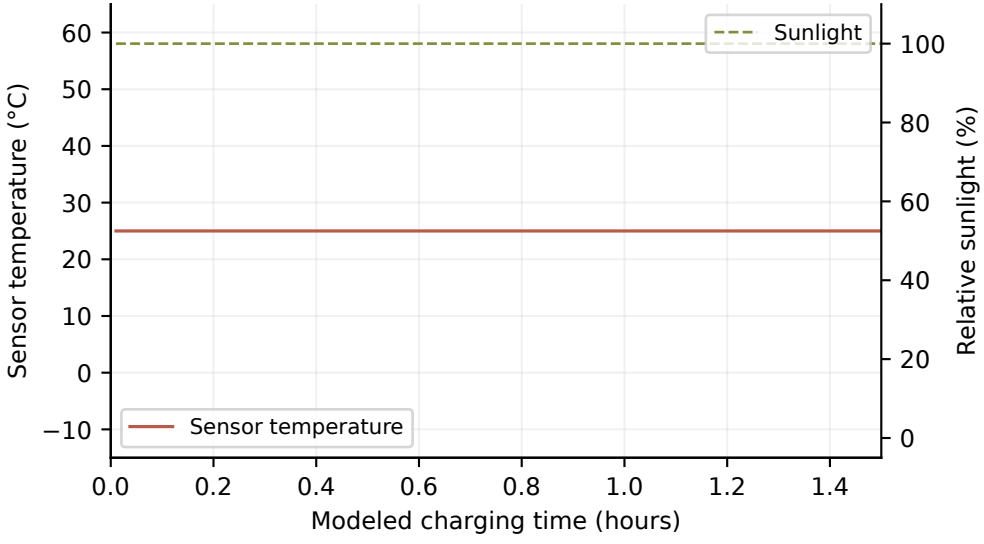
An approximately 2.85 V model cell starts at 75 mA preconditioning, rises above the threshold, and moves to 500 mA charging. This is not permission to recover a damaged real cell.

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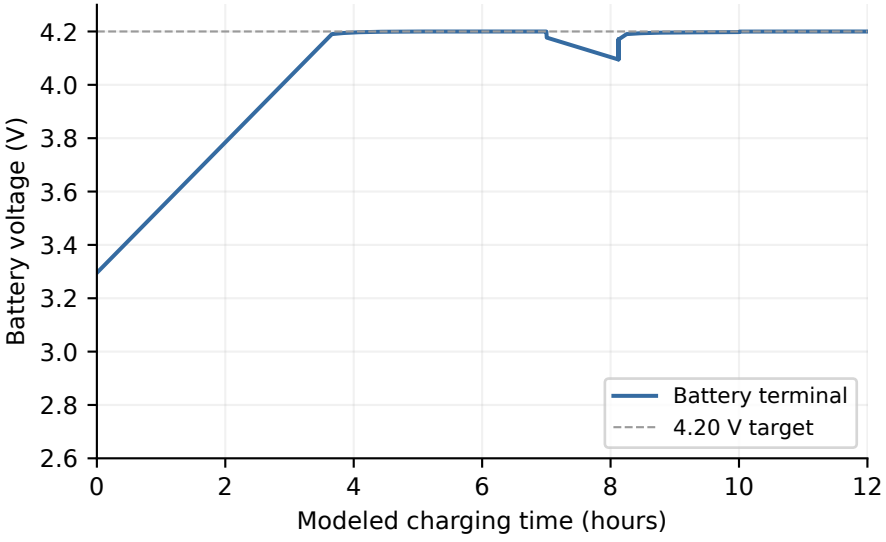


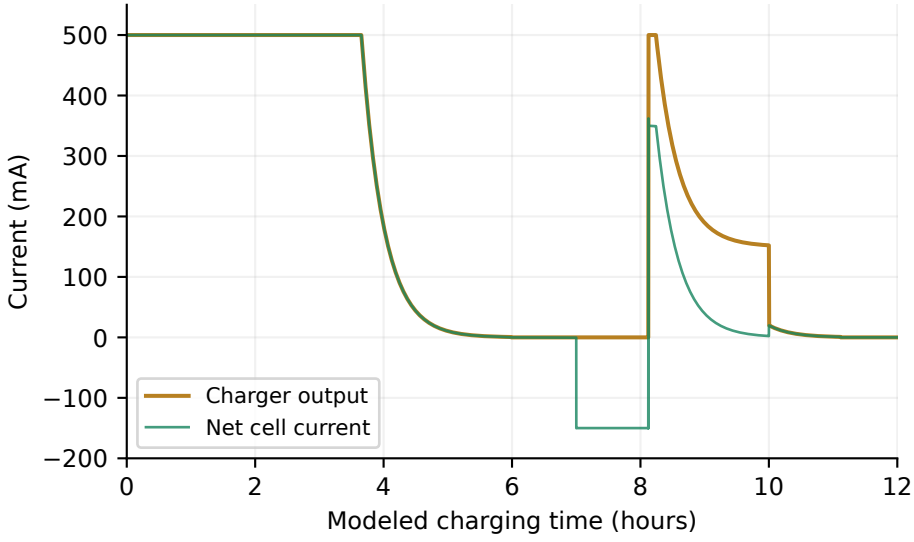
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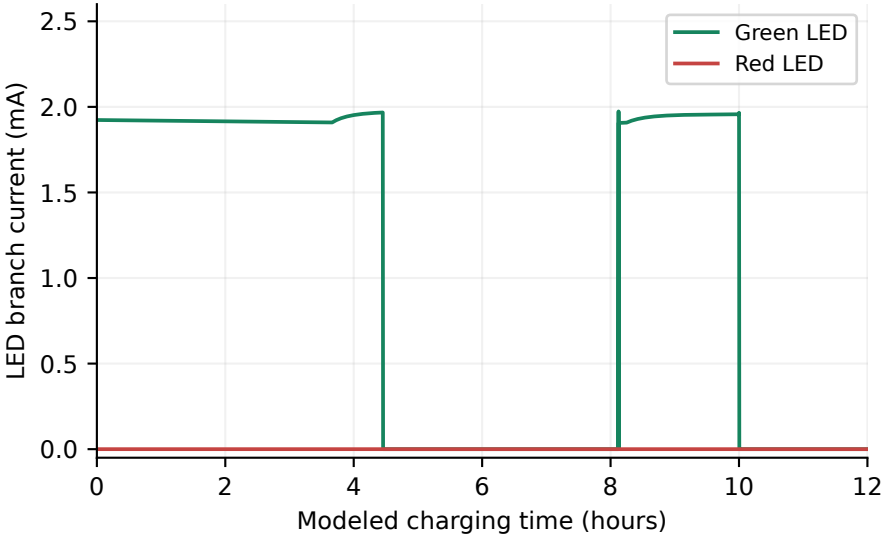
06 Full battery, load, and recharge

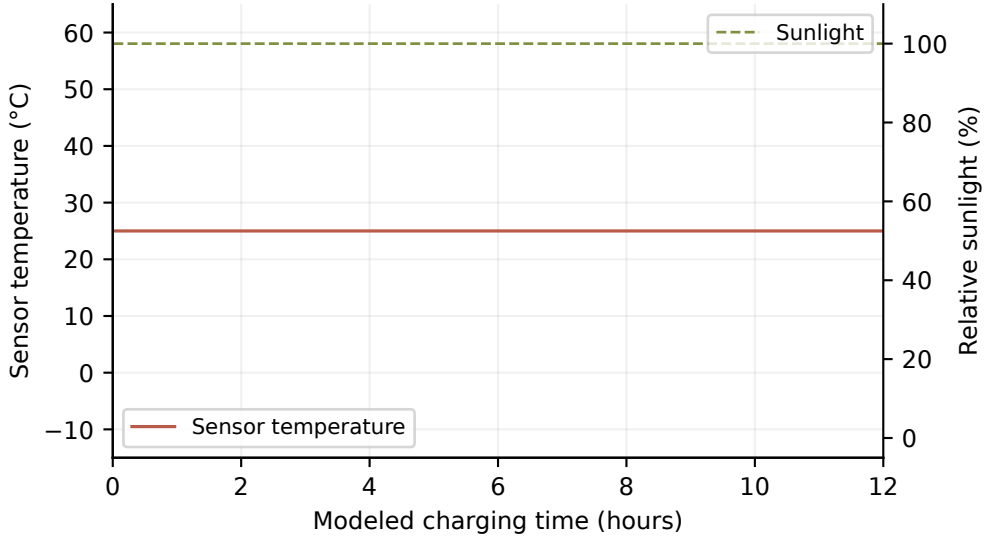
After timer termination a 150 mA external load discharges the cell. Below the recharge threshold the charger restarts. An external load can affect charging and termination.

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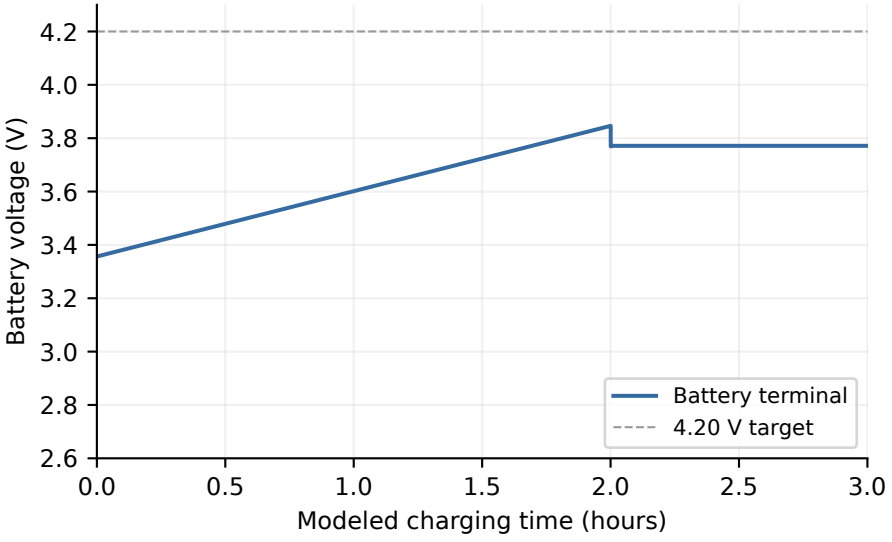


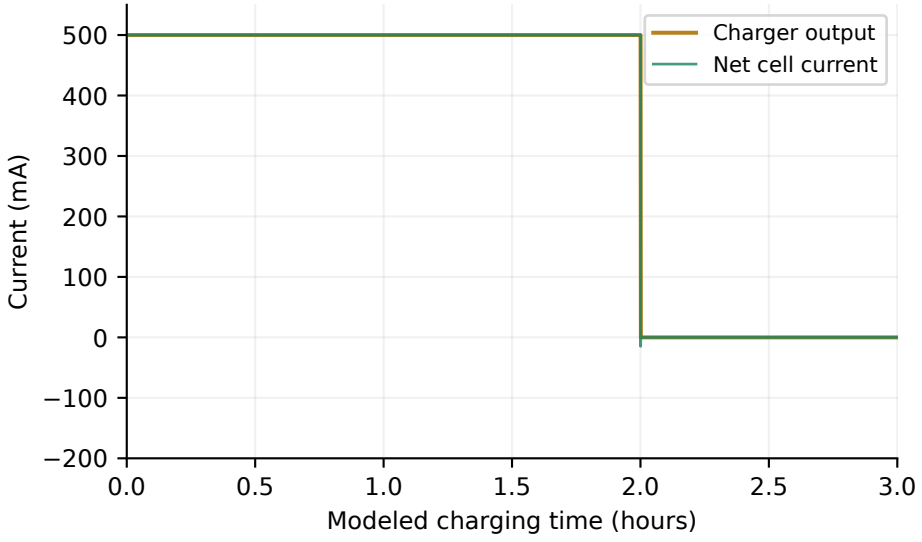
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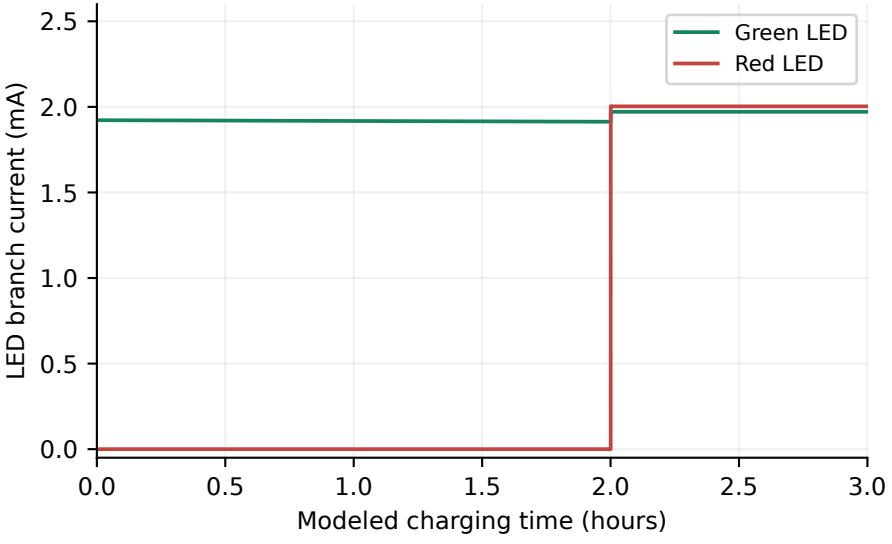
07 Disconnected temperature sensor

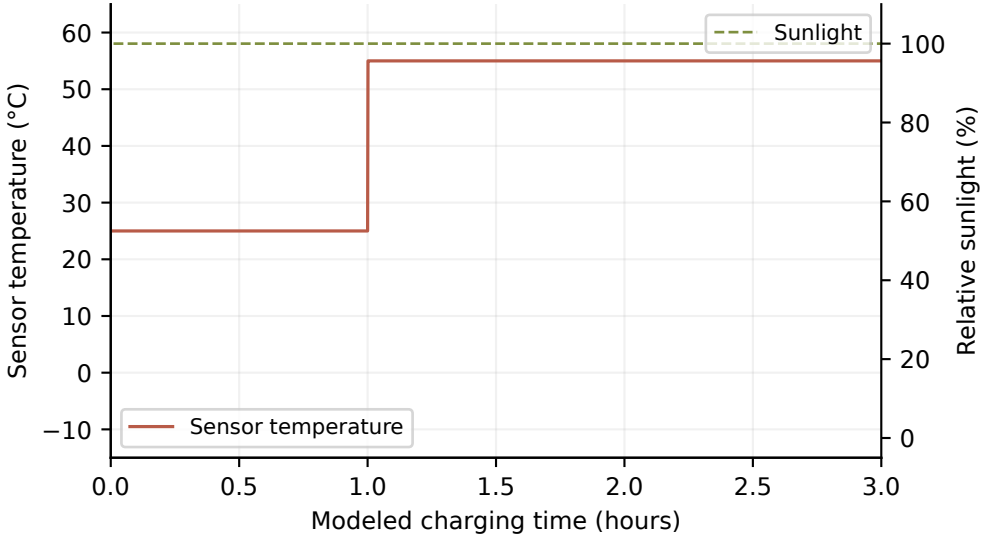
Opening the NTC disables temperature monitoring; it does not produce an open-sensor alarm. Reconnecting it at 55 C produces the temperature fault.

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